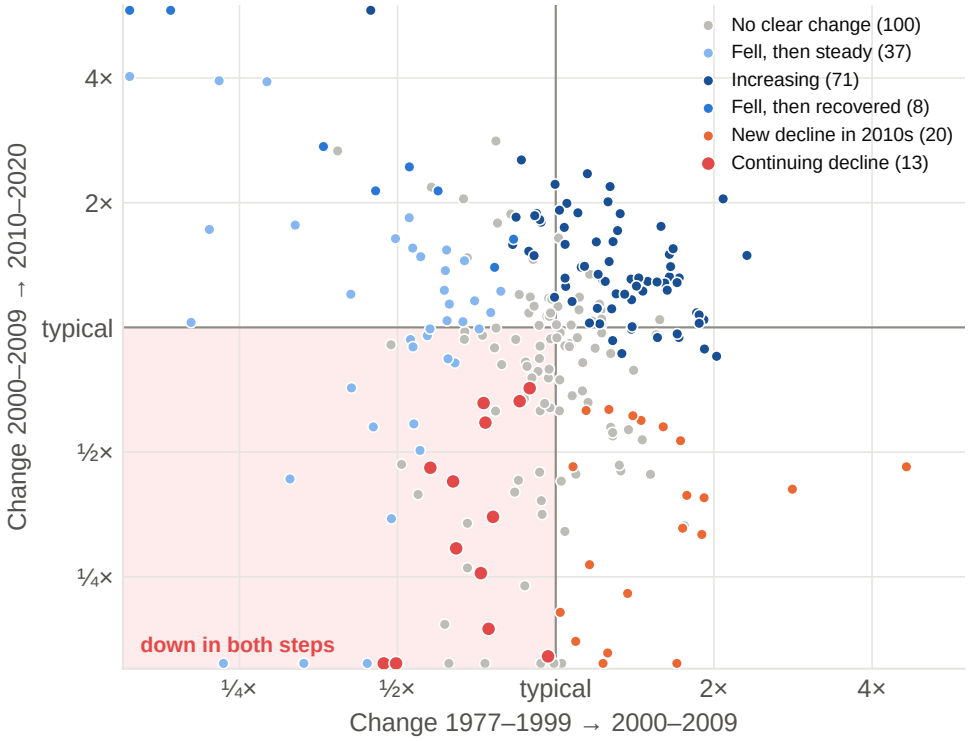
QUESTION 1

Which species, if any, are continuing to decline in breeding?

13 native species are still breeding less, decade after decade

Breeding share vs typical species: 1977–1999 → 2000–2009 (across), then → 2010–2020 (up)



Still declining (change 1977→2020)

Intermediate Egret (colonial waterbird)	-97%
Straw-necked Ibis (colonial waterbird)	-94%
Great Egret (colonial waterbird)	-85%
Fuscous Honeyeater	-84%
Nankeen Night-Heron (colonial waterbird)	-81%
Bar-shouldered Dove	-80%
Australian Reed-Warbler	-72%
Little Egret (colonial waterbird)	-72%
White-throated Gerygone	-72%
Cattle Egret (colonial waterbird)	-55%
Yellow-rumped Thornbill	-50%
Olive-backed Oriole	-41%
Black-faced Cuckoo-shrike	-33%

KEY FINDINGS

- 13 of 249 native species bred less in both 2000–2009 and 2010–2020, compared with the typical species: 6 colonial waterbirds (Intermediate Egret, Straw-necked Ibis, Great Egret, Nankeen Night-Heron) and 7 other species, mostly woodland birds (Fuscous Honeyeater, Bar-shouldered Dove, Australian Reed-Warbler, White-throated Gerygone).
- 8 species fell in 2000–2009 and recovered in 2010–2020 (e.g. Masked Woodswallow, Black-tailed Native-hen, Beach Stone-curlew); 37 fell and then held steady.
- 71 species increased; most species (100) show no clear change.

POSSIBLE CAUSES · OUR SPECULATION, NOT TESTED HERE

- Colonial waterbirds breed in a few big inland colonies when rivers flood. Their breeding stayed low in 2010–2020 despite the 2010–12 floods, so less water reaching colony wetlands (river regulation and extraction), or fewer Atlas visits to colonies, may be the cause. Testable against aerial waterbird colony counts.
- Drought alone does not fit the continuing decliners: the species that recovered after the Millennium Drought ended show what a drought response looks like, and these did not recover.
- Noisy Miners are unlikely to explain the woodland birds: in 'Hidden Pressures', their breeding odds with miners were 0.90–1.13× (about the same). Loss of shrub and nest sites in ageing woodland remnants is a more likely candidate. Prediction: these species keep declining in DES 2021–25.

Periods: 1977–1999 is the team's agreed early period; the recent period is split at 2010 to test whether declines continued after the Millennium Drought. Breeding share = sightings with breeding evidence, Atlas surveys only, same month mix, vs the typical species. Several decliners have under 10 breeding records in 2010–2020. Table: outputs/Q1_breeding_trends.csv